Olympics Analytics Dashboard

SQL-Powered Analysis · Historical 1896–2024 · Plotly Dash

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20,387 TOTAL MEDALS	**14,948** ATHLETES
165 COUNTRIES	**38** OLYMPIC EDITIONS

Project Overview

The Problem

- 128 years of Olympic data scattered
- across two Kaggle datasets
- No unified view of historical trends
- No interactive way to slice and filter
- Raw CSVs with inconsistent formatting

The Approach

- SQL-only data cleaning via CREATE VIEW
- Deduplication with ROW_NUMBER()
- UNION ALL to merge both datasets
- Parameterized Dash callbacks
- Zero Python in the cleaning layer

The Outcome

- 20,387 medals unified in one view
- 9 interactive charts + 5 filters
- Paris 2024 spotlight section
- Real-time SQL filtering on every query
- Secure, auditable, reproducible

Tech Stack & Architecture

Kaggle API

2 datasets downloaded

SQLite DB

All CSVs loaded

SQL Views

Clean, deduplicate

UNION ALL

all_olympics view

Plotly Dash

9 charts
5 filters

Key SQL Techniques

- ROW_NUMBER() OVER (PARTITION BY ...) — deduplication
- SUBSTR(slug_game, -4) — year extraction from slugs
- COALESCE + UPPER + TRIM — standardisation
- UNION ALL — merging historical + Paris 2024
- Parameterized ? — SQL injection prevention

Three Views Power Everything

`clean_historical`	Deduped, standardised 1896–2022 medals
`clean_paris2024`	Cleaned Paris 2024 with matching schema
`all_olympics`	UNION of both — 20,387 total records

Dashboard · KPIs & Medals Over Time

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TOTAL MEDALS	ATHLETES	COUNTRIES	SPORTS	EDITIONS	GOLD MEDALS	SILVER MEDALS	BRONZE MEDALS
20,387	14,948	165	76	38	6,639	6,605	7,143
All-time	Unique	Participated	Disciplines	Games held			

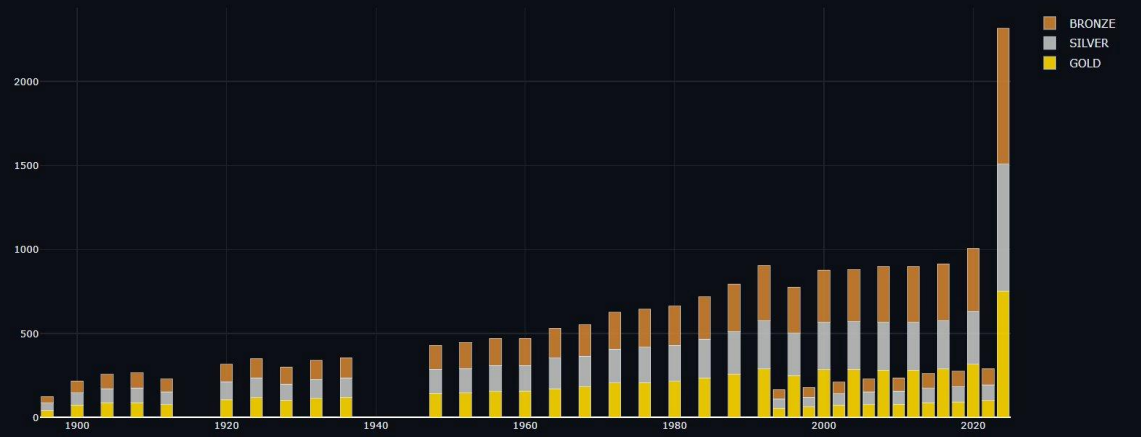
Year Range: 1896 – 2024 (1896, 1916, 1936, 1956, 1976, 1996, 2016)

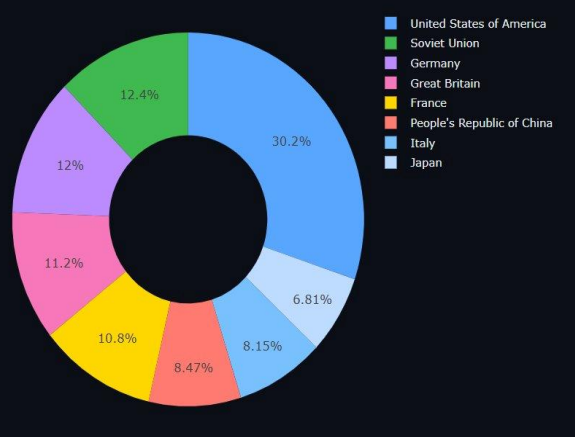
Country: All Countries

Sport: All Sports

Medal: All Medals

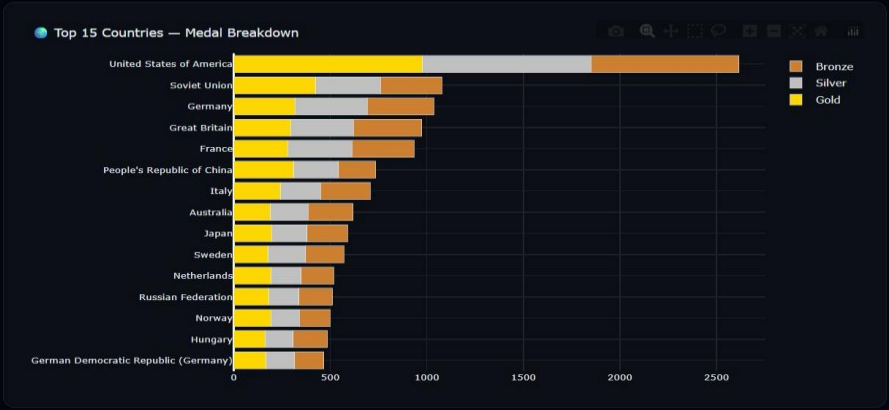
Season: All Seasons

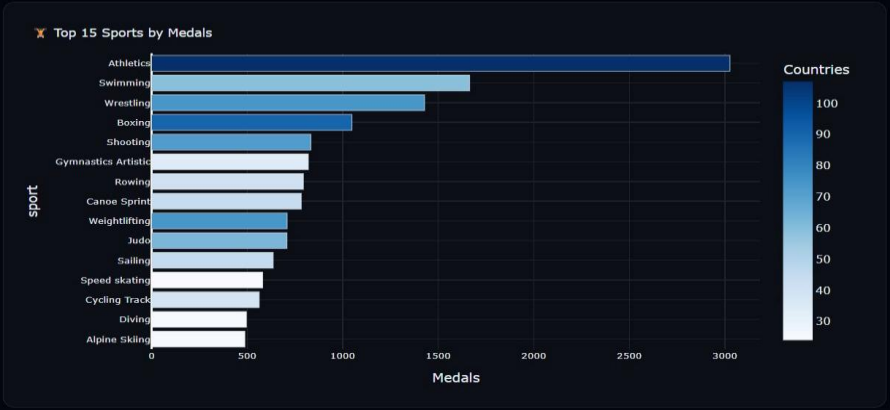


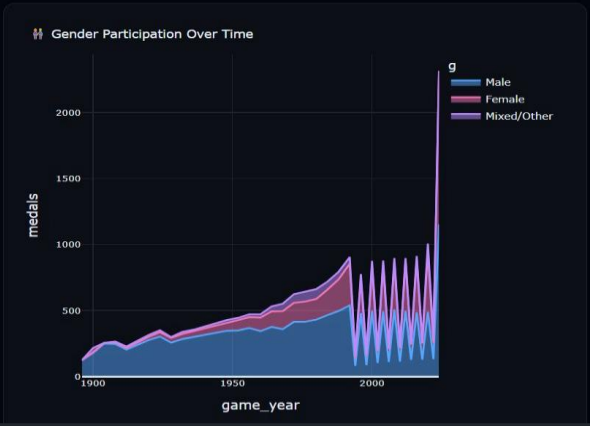


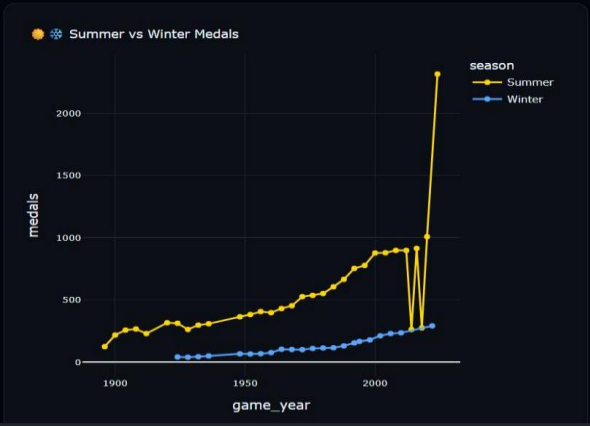
Paris 2024 spike visible — largest medal tally in Olympic history with 2,000+ medals awarded across 45 disciplines

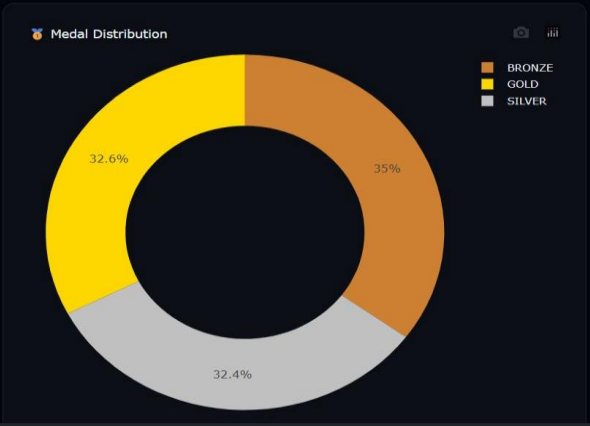
Dashboard · Country & Sport Analysis







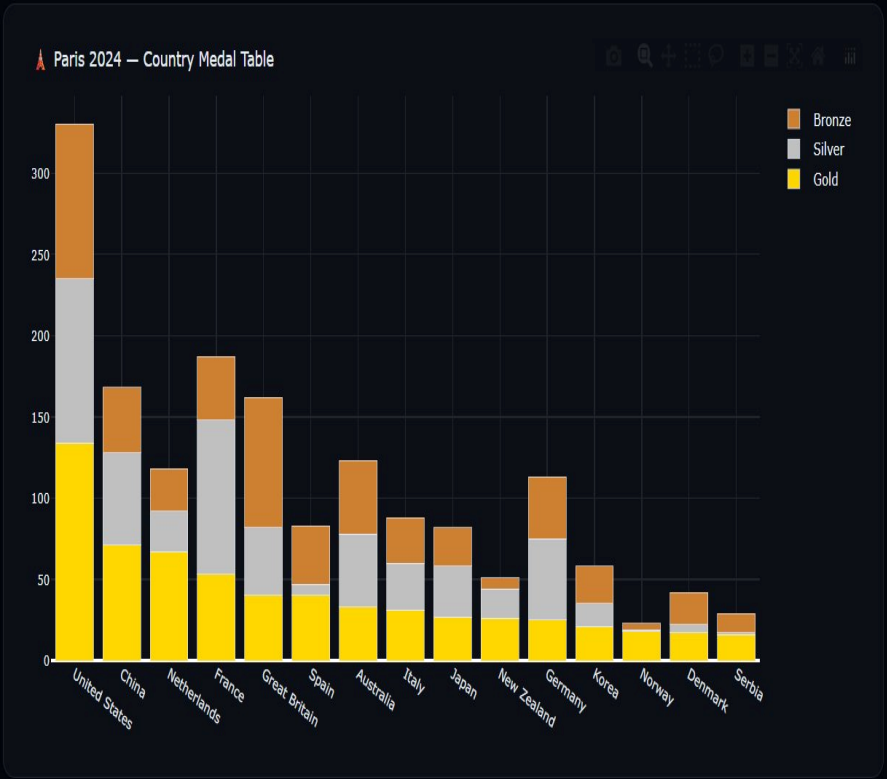


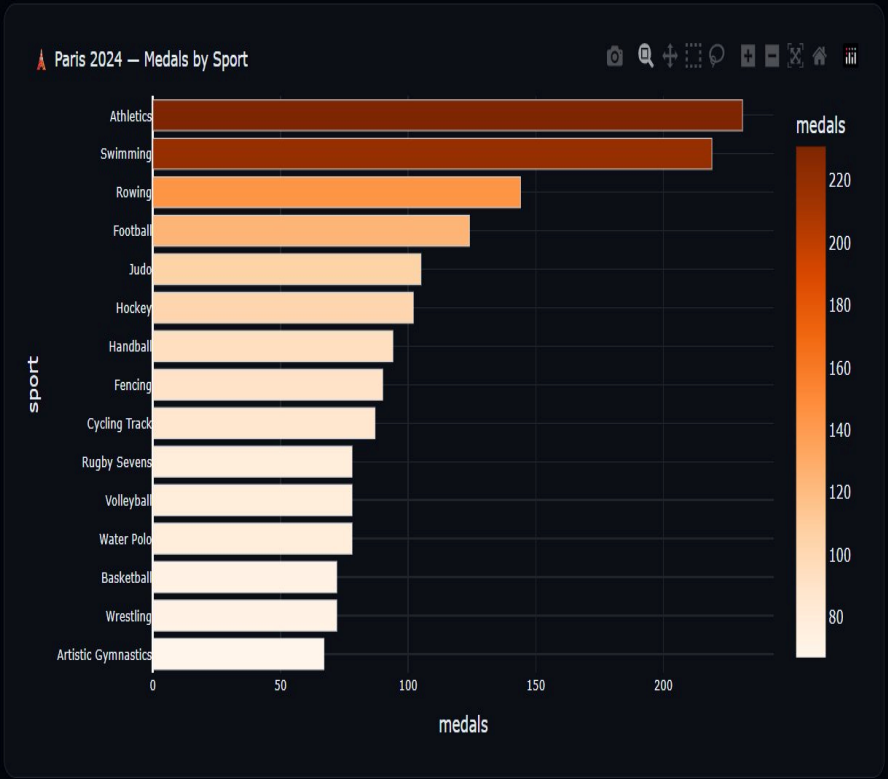


USA leads all-time with 2,600+ medals · Athletics dominates sports · Female participation rose sharply post-2000

Dashboard · Paris 2024 Spotlight

Paris 2024 Spotlight





USA led Paris 2024 · Athletics & Swimming most competitive · Netherlands surprised with top-3 gold finish

Key Insights

📈 Participation Growth

From 14 nations in 1896 to 165 countries in 2024. Medal count grew from 43 to 2,000+ per Games — driven by new sports and expanded events.

🙂 Gender Inclusion

Female medals were negligible pre-1950. Post-2000, female participation surged sharply — now nearly equal to male medal counts at major Games.

US Dominant Nations

USA holds 30% of all-time medals among the top 8. Soviet Union dominated Cold War era. China emerged strongly post-1984. Netherlands surprised at Paris 2024.

🏃 Sport Dominance

Athletics and Swimming account for the most medals historically. Color-coded by country diversity — Athletics has highest spread across nations.

❄️ Summer vs Winter

Summer Games consistently produce 3x–4x more medals than Winter. Both grew steadily. Paris 2024 shows highest Summer spike ever recorded.

🗼 Paris 2024 Standouts

USA led overall. Netherlands punched above their weight in Gold medals. Athletics and Swimming were the most contested disciplines across 45 sports.

Engineering Decisions

Why SQL-only cleaning?

Keeps the transformation layer declarative and auditable. Any analyst can read a CREATE VIEW statement. Python transformations are harder to audit and often mix business logic with display logic.

Why in-memory SQLite?

For a demo dashboard, in-memory keeps setup stateless and simple — no database server, no migrations. The trade-off is rebuild time on each run, which is acceptable for this dataset size (~20K rows).

Why parameterized queries?

Parameterized queries (? placeholders) are non-negotiable in production. String interpolation in SQL is a SQL injection vector. Every Dash callback uses params= to pass values safely.

Why Plotly Dash over Streamlit?

Dash gives full control over layout, styling, and callback logic. The dark theme and custom CSS required for this design aren't easily achievable in Streamlit without significant workarounds.

Thank You

SQL · SQLite · Plotly Dash · Kaggle · Python

20,387	128	9	100%
Medals Analysed	Years of Data	Interactive Charts	SQL-Cleaned

github.com/patilgaurang0907